

Off-Road Semantic Segmentation using U-Net

Team Name: Bharath Kumar Alla

Tagline: Enhancing terrain understanding for autonomous navigation using deep learning

Introduction

Semantic segmentation plays a crucial role in enabling machines to understand complex environments by assigning a class label to every pixel in an image. In the context of autonomous navigation, especially in off-road environments, it becomes essential to accurately distinguish between terrain types such as dirt, vegetation, obstacles, and other surface conditions.

This project focuses on developing a deep learning-based semantic segmentation system using the U-Net architecture. The primary objective is to improve terrain understanding, which can be further leveraged in applications such as self-driving vehicles, robotics, and intelligent navigation systems. The proposed approach aims to achieve high segmentation accuracy while maintaining computational efficiency.

The system is implemented using PyTorch and trained on an off-road segmentation dataset. The workflow includes data preprocessing, model design, training, and evaluation using standard metrics such as Intersection over Union (IoU).

Methodology

The methodology adopted in this project involves a structured pipeline starting from dataset preparation to model evaluation. The dataset is initially extracted and organized into training and validation subsets. Each image is paired with a corresponding segmentation mask that provides pixel-level annotations.

To ensure uniformity and reduce computational overhead, all images and masks are resized to a resolution of 256x256 pixels. Image normalization is performed by scaling pixel values between 0 and 1, which stabilizes the training process and accelerates convergence.

A custom dataset class is implemented using PyTorch to handle efficient data loading and preprocessing. This class reads images and masks, applies necessary transformations, and converts them into tensor format suitable for model input.

One of the key preprocessing steps involves label mapping. The original dataset contains irregular class labels, which are remapped into a fixed set of six classes. This ensures consistency across the dataset and simplifies the learning process for the model.

The processed data is then fed into a DataLoader, which enables batching and shuffling. This improves training efficiency and ensures better generalization of the model.

Model Architecture and Training

The core of this project is the U-Net architecture, a convolutional neural network specifically designed for semantic segmentation tasks. It consists of an encoder-decoder structure that allows the model to capture both contextual and spatial information.

The encoder path is responsible for extracting features from the input image. It uses convolutional layers followed by activation functions and pooling operations to progressively reduce spatial dimensions while increasing feature depth.

At the bottleneck, the network captures high-level abstract representations of the input. This helps the model understand complex patterns and relationships within the data.

The decoder path performs upsampling using transpose convolutions and combines feature maps from the encoder through skip connections. These skip connections play a crucial role in preserving fine-grained spatial details that are often lost during downsampling.

The final layer outputs a multi-class segmentation map corresponding to the predefined terrain classes.

For training, the Adam optimizer is used with a learning rate of 0.001. The loss function employed is CrossEntropyLoss, which is well-suited for multi-class classification problems. The model is trained over multiple epochs, with performance monitored using validation data.

Results, Challenges, and Conclusion

The performance of the model is evaluated using the Intersection over Union (IoU) metric. IoU measures the overlap between predicted segmentation and ground truth, providing a robust evaluation of model accuracy.

During training, the model shows a consistent decrease in loss and an improvement in IoU scores across epochs. This indicates effective learning and convergence of the model.

However, several challenges were encountered during the development process. One major issue was label inconsistency in the dataset, which was addressed through label mapping. Another challenge was the high computational cost associated with large image sizes, which was mitigated by resizing images.

Memory limitations in the training environment were handled by reducing batch size and optimizing data loading mechanisms. Additionally, class imbalance in the dataset affected performance, which was addressed by focusing on IoU as the primary evaluation metric.

In conclusion, the project successfully demonstrates the application of deep learning for semantic segmentation in off-road environments. The U-Net model effectively captures both global context and fine details, making it suitable for terrain understanding tasks.

Future work can focus on integrating more advanced architectures such as ResNet-based encoders, applying data augmentation techniques, and exploring alternative loss functions like Dice Loss and Focal Loss. Further improvements can also include real-time deployment and optimization for edge devices.